Block practice · Storage devices, utility software, cyber attacks and Pigpen

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Cyber attacks: phishing, shoulder surfing, eavesdropping and unpatched software

Utility software: anti-malware, defrag and file tools

How hard drives, optical discs and SSDs store data

OLevels.com SCHOOL

Practice workbook · Cyber attacks: phishing, shoulder surfing, eavesdropping and unpatched software

Defining and recognising attacks, spotting phishing indicators in an email, and explaining how an attack works or is prevented.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T56-09

1. Question 2 · part c.i · 2 marks

2021 NOV · Paper 01

Worked-example retry · Attempt part c.i only.

2 Danielle manages network security for a small business.

(a) Identify **one** action Danielle could take to protect the **network** from unauthorised access.

(1)

- ☒ **A** Give each user a separate account
- ☒ **B** Enforce a strong password policy
- ☒ **C** Make a backup plan
- ☒ **D** Encrypt sensitive files

(b) Danielle has received this email.

She is a customer of SafePayment.com but thinks this is a phishing attempt.

From: SafePayment.accounts@5afepayment.com
To: Danielle616
Subject: Account restricted (case SP-0011312-2021-06)

Dear Customer,

We have noticed some unusual activity on your account, so have stopped all payments.

Please use this link to log in and check your account.

Log in

Your normal log in will stoop working until you have done this.

Regards
SafePayment fraud prevention team.

(i) Danielle has circled the words **Dear Customer**.

Give **one** reason why the use of these words might make Danielle suspect that the email is a phishing attempt.

(1)

(ii) Draw circles around **two other** aspects of the email that might make Danielle suspect it is a phishing attempt.

(2)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

6

P 6 9 2 9 7 A 0 6 2 0

1. Question 2 · part c.i · 2 marks

2021 NOV · Paper 01

Worked-example retry · Attempt part c.i only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(c) Some employees use company laptops in public places.

Danielle is concerned that shoulder surfing could pose a risk to security.

(i) Describe what is meant by **shoulder surfing**.

(2)

(ii) Explain **one** way to prevent shoulder surfing.

(2)

OLevels.COM SCHOOL

(d) Danielle wants to protect the network from cyberattacks.

(i) Identify the purpose of penetration testing.

(1)

- ☒ **A** To decompile executable code
- ☒ **B** To identify security vulnerabilities
- ☒ **C** To test the strength of an encryption algorithm
- ☒ **D** To enforce a network user policy

P 6 9 2 9 7 A 0 7 2 0

7

Turn over ▶

1. Question 2 · part c.i · 2 marks · Working and answer

Working and answer

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2. Question 2 · part b.i · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part b.i only.

2 Danielle manages network security for a small business.

(a) Identify **one** action Danielle could take to protect the **network** from unauthorised access.

(1)

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- ☒ **C** Make a backup plan
- ☒ **D** Encrypt sensitive files

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(1)

.....

.....

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(2)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

6

P 6 9 2 9 7 A 0 6 2 0

2. Question 2 · part b.i · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 2 · part b.ii · 2 marks

2021 NOV · Paper 01

Further practice · Attempt part b.ii only.

2 Danielle manages network security for a small business.

(a) Identify **one** action Danielle could take to protect the **network** from unauthorised access.

(1)

- ☒ **A** Give each user a separate account
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.....

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DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

6

P 6 9 2 9 7 A 0 6 2 0

3. Question 2 · part b.ii · 2 marks

2021 NOV · Paper 01

Further practice · Attempt part b.ii only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(c) Some employees use company laptops in public places.

Danielle is concerned that shoulder surfing could pose a risk to security.

(i) Describe what is meant by **shoulder surfing**.

(2)

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(2)

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- ☒ **A** To decompile executable code
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3. Question 2 · part b.ii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 2 · part c.ii · 2 marks

2021 NOV · Paper 01

Further practice · Attempt part c.ii only.

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P 6 9 2 9 7 A 0 6 2 0

4. Question 2 · part c.ii · 2 marks

2021 NOV · Paper 01

Further practice · Attempt part c.ii only.

DO NOT WRITE IN THIS AREA

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(c) Some employees use company laptops in public places.

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(1)

- ☒ **A** To decompile executable code
- ☒ **B** To identify security vulnerabilities
- ☒ **C** To test the strength of an encryption algorithm
- ☒ **D** To enforce a network user policy

4. Question 2 · part c.ii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 2 · part d.ii · 2 marks

2021 NOV · Paper 01

Further practice · Attempt part d.ii only.

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(a) Identify **one** action Danielle could take to protect the **network** from unauthorised access.

(1)

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DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

5. Question 2 · part d.ii · 2 marks

2021 NOV · Paper 01

Further practice · Attempt part d.ii only.

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(2)

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- ☒ **C** To test the strength of an encryption algorithm
- ☒ **D** To enforce a network user policy

5. Question 2 · part d.ii · 2 marks

2021 NOV · Paper 01

Further practice · Attempt part d.ii only.

(ii) Danielle is sent a software patch for the network operating system but is too busy to install it straight away.

Explain why this delay could pose a threat to the security of the network.

(2)

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(Total for Question 2 = 11 marks)

DO NOT WRITE IN THIS AREA

Working and answer

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6. Question 5 · part c · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part c only.

5 Viza Health Centre is located in the North East of England.

(a) The health centre uses artificial intelligence to provide a symptom-checking service for its patients.

Patients log on to the website and input their symptoms.

(i) Describe how artificial intelligence could identify what is wrong with them.

(2)

(ii) Give **one** reason why a patient may not want to use this online service.

(1)

(b) The health centre has clinics in two buildings: Cleveland and Stockton.

The network server is in the Cleveland building.

(i) Name the type of network used to access the server from within the Cleveland building.

(1)

(ii) Name the type of network used to access the server from the Stockton building.

(1)

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6. Question 5 · part c · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part c only.

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(c) The network is at risk of an eavesdropping attack.

Identify the description of eavesdropping.

(1)

- ☒ **A** Tricking people into giving information by sending emails pretending to be from someone in authority
- ☒ **B** Spying on someone using a computer
- ☒ **C** Intercepting information as it is transmitted over a network
- ☒ **D** Redirecting a user from a genuine website to a fake one

(d) Doctors use laptops when they visit patients in their homes.

(i) The laptops have solid state drives.

Explain **one** reason why a solid state drive is better than a magnetic hard drive for the laptops.

(2)

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(ii) Describe how data is stored on a solid state drive.

(2)

(iii) The laptops have two types of memory.

Complete the table by adding **one** tick (✓) to match **each** description to the type of memory used.

(2)

Description	RAM	ROM
Stores the boot up sequence		
The contents are lost when the laptop is shut down		

(Total for Question 5 = 12 marks)

6. Question 5 · part c · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 1 · part b.iii · 1 mark

2023 JUN · Paper 01

Further practice · Attempt part b.iii only.

1 People use networks in all aspects of their lives.

(a) An office building has desktop computers connected in a network.

One reason for networking computers is to enable technicians to maintain them remotely.

State **two other** reasons for connecting computers in a network.

(2)

1 .....

.....

2 .....

.....

(b) Access to data must be controlled.

(i) Identify **one** reason that encryption is used.

(1)

☒ **A** To compress the file so it takes up less storage space

☒ **B** To make sure both sender and receiver use the same key

☒ **C** To make sure data is only understood by the intended receiver

☒ **D** To stop malicious hackers getting into a computer network

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7. Question 1 · part b.iii · 1 mark

2023 JUN · Paper 01

Further practice · Attempt part b.iii only.

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(ii) File servers are in a small room at the back of an office.

The office has a burglar alarm.

Give **two other** ways that the servers can be secured using physical methods.

(2)

1 .....

.....

2 .....

.....

(iii) Phishing is a type of social engineering.

State what is meant by the term **social engineering**.

(1)

.....

.....

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(c) 5G is a communication standard.

Identify the characteristic that is true for the 5G communication standard.

(1)

- ☒ **A** It has a high transmission latency
- ☒ **B** It has built-in security
- ☒ **C** It has a lower bandwidth than 3G and 4G
- ☒ **D** It uses wireless communication

7. Question 1 · part b.iii · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Utility software: anti-malware, defrag and file tools

Purpose and functions of utilities: anti-malware types and features, defragmentation, backup, repair and conversion, including choosing a utility for an issue and showing a disc after defragmenting.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: CSC-T4-13

1. Question 5 · part c · 4 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part c only.

5 A team is developing a computer game.

(a) Complete the table by adding a tick (✓) to show the characteristics of program language translators.

You may select more than one translator per characteristic.

(3)

Characteristic	Translator		
	Assembler	**Interpreter**	**Compiler**
Does **not** have to be present in memory to execute the translated code.			
Translates high level language source code.			
Executes high level code until either the program has finished, or a syntax error is encountered.			
Translates low level language source code into machine code.			

(b) Luca is testing the computer game.

The computer that he is using is an older model with a processor that runs at a low clock speed.

Explain **one** drawback of testing the game using this computer.

(2)

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1. Question 5 · part c · 4 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part c only.

(c) Anna uses utility software.

Complete the table by adding the type of utility software she should use to solve each issue.

Issue	Utility software
Her word processor cannot read all of a file because it is corrupt.	
She has a JPEG image and needs to send it to a publisher who can only read PNG image files.	
There has been a flood and files on a server have been damaged, but Anna needs to access her data.	
Her hard drive is running very slowly when she tries to access a file.	

(4)

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OLevels.com SCHOOL

1. Question 5 · part c · 4 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part c only.

(d) Tarik is writing an instruction manual for the computer game.

He decides to use software to compress the instruction manual before emailing it to a proofreader for checking.

State and justify the most appropriate compression method.

(3)

Method

.....

Justification

.....

.....

.....

.....

(e) Members of the team developing the game are either coders or designers.

Coders (C) write the application code.

Designers (D) design the graphics for the game and write instruction manuals.

Team members have one role only.

Construct a logic statement, using AND, OR and NOT with the letters C and D, to show the conditions for being a member of the team.

(3)

(Total for Question 5 = 15 marks)

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1. Question 5 · part c · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 3 · part e · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part e only.

3 Zakir wants to buy a new laptop.

(a) He knows that a laptop has both hardware and software.

Explain why the laptop needs both hardware and software.

(2)

(b) Zakir needs to choose between a solid state drive and a hard disc drive.

(i) Give **two** reasons, other than cost and capacity, why Zakir may choose a solid state drive rather than a hard disc drive.

(2)

1

2

(ii) Solid state drives store data on flash memory, which is similar to RAM.

Explain **one** difference between flash memory and RAM.

(2)

(c) One of the laptops has 4 megabytes (MB) of cache memory.

Explain the purpose of cache memory.

(2)

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OLevels.com SCHOOL

2. Question 3 · part e · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part e only.

DO NOT WRITE IN THIS AREA

(d) Zakir uses many devices.

Complete the table by adding **one** tick (✓) in each row to show whether the device is an input device, an output device or both.

(2)

Device	Input	Output	Both
Headset			
Mouse			
Printer			
Touch screen			

(e) Zakir needs to protect his laptop against malware.

Give **two** types of anti-malware software he could install.

(2)

1 .....

2 .....

(Total for Question 3 = 12 marks)

DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

Working and answer

3. Question 6 · part b.i · 2 marks

2021 NOV · Paper 01

Further practice · Attempt part b.i only.

6 Akiko develops software programs.

(a) Akiko can use a low-level or a high-level programming language to write code.

(i) Give **one** benefit of using a low-level programming language to develop a program.

(1)

(ii) Give **one** drawback of using a low-level programming language to develop a program.

(1)

(b) Akiko is writing an anti-virus program.

(i) The program will include a virus removal function.

Give **two other** functions that should be included.

(2)

1 .....

2 .....

(ii) An anti-virus program is one type of anti-malware.

Give **one other** type of anti-malware.

(1)

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OLevels.COM SCHOOL

18

P 6 9 2 9 7 A 0 1 8 2 0

3. Question 6 · part b.i · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 6 · part b.ii · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part b.ii only.

6 Akiko develops software programs.

(a) Akiko can use a low-level or a high-level programming language to write code.

(i) Give **one** benefit of using a low-level programming language to develop a program.

(1)

(ii) Give **one** drawback of using a low-level programming language to develop a program.

(1)

(b) Akiko is writing an anti-virus program.

(i) The program will include a virus removal function.

Give **two other** functions that should be included.

(2)

1 .....

2 .....

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Give **one other** type of anti-malware.

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OLevels.COM SCHOOL

4. Question 6 · part b.ii · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part b.ii only.

(c) Akiko is also working on a burglar alarm system.

The alarm must only be triggered if:

- the alarm (A) is set
- it is dark outside (O)
- a window (W) or a door (D) has been opened.

Construct a Boolean expression, using AND, OR and NOT with the letters A, O, W, and D to show the conditions that will trigger the alarm.

(4)

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4. Question 6 · part b.ii · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 3 · part d · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part d only.

3 Software programs carry out tasks on computers.

(a) A headteacher uses a computer simulation to model the effect of increasing class sizes.

Explain **one** problem with using simulations to predict the effects of changes.

(2)

(b) Images are stored as files before being displayed or printed.

(i) Give **one** measurement of image resolution.

(1)

(ii) Increasing the colour depth of an image leads to an increase in the image file size.

Describe the reason for the increase in file size.

(2)

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5. Question 3 · part d · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part d only.

(c) Software is divided into two categories.

Describe **one** difference between system software and application software.

(2)

(d) Here is an image of secondary storage.

Two files (W and Z) are stored on it.

Each file is made up of several blocks (e.g. Z1, Z2, Z3).

	W4	Z1				W2		W3	Z3		W1		Z2

Complete the image to show the state after running a defragmentation utility.

(2)

DO NOT WRITE IN THIS AREA

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DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

P 7 2 9 3 7 A 0 9 2 4

5. Question 3 · part d · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part d only.

(e) Here is part of a file that contains electric meter readings.

04631 04984 05103 05163 05271 05383 05487 05722

Explain the effect on the file of applying a run-length encoding algorithm to this data file.

(2)

.....

.....

.....

.....

(f) A series of tasks is written in different programming languages.

Complete the table to show the correct programming language translator for each task.

(4)

Task	Translator
A guessing game that can be used on different computing platforms	
A screen driver for a new smartphone	
A new version of a spreadsheet program for sale next year	
Control software for an embedded system inside a new washing machine	

(Total for Question 3 = 15 marks)

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OLevels.com SCHOOL

5. Question 3 · part d · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · How hard drives, optical discs and SSDs store data

How data is physically stored and read on magnetic, optical and solid state devices, and reasons to choose a solid state drive over a hard disc.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

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OLevels.com SCHOOL

Type ID: CSC-T4-07

1. Question 2 · part b.ii · 3 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part b.ii only.

2 Computer systems use hardware and software.

(a) Two types of software are application software and system software.

(i) State the purpose of application software.

(1)

.....

.....

(ii) One function of an operating system is to provide a user interface.

State **three** other functions of an operating system.

(3)

1 .....

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2 .....

.....

3 .....

.....

OLevels.com SCHOOL

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

1. Question 2 · part b.ii · 3 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part b.ii only.

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DO NOT WRITE IN THIS AREA

(b) Computers use different types of storage.

(i) Explain **one** reason why computers need both RAM and ROM.

(2)

(ii) Describe how data is read from a magnetic hard drive.

(3)

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(c) Computers perform the fetch-decode-execute cycle.

Complete the table by writing the numbers 1 to 4 in the sequence number column to put each stage in the correct order.

Stage	Sequence number
The instruction is placed onto the data bus	
The Arithmetic and Logic Unit (ALU) performs a calculation	
The address of the next instruction is placed onto the address bus	
The instruction is decoded by the control unit	

(4)

1. Question 2 · part b.ii · 3 marks · Working and answer

Working and answer

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2. Question 1 · part e · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part e only.

1 Computers are made up of both hardware and software components.

(a) Computers use a three-step computational model.

Complete the diagram with the names of the **three** steps.

(3)



(b) Identify the hardware component responsible for controlling the fetch-decode-execute cycle.

(1)

- ☒ **A** Cache
- ☒ **B** Central processing unit
- ☒ **C** ROM
- ☒ **D** Address bus

(c) Von Neumann developed the stored program concept that permits two different types of item to reside in memory.

Name these **two** types.

(2)

1 .....

.....

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2. Question 1 · part e · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part e only.

(d) Some computers use virtual memory.

Explain how virtual memory works.

(2)

(e) An optical storage device writes data onto a CD or DVD.

Describe how data is stored physically on optical media.

(2)

(f) A disc-formatting program is an example of a type of software.

Identify the type.

(1)

- ☒ **A** Security software
- ☒ **B** Application software
- ☒ **C** Translation software
- ☒ **D** System software

(Total for Question 1 = 11 marks)

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2. Question 1 · part e · 2 marks · Working and answer

Working and answer

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3. Question 3 · part b.i · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part b.i only.

3 Zakir wants to buy a new laptop.

(a) He knows that a laptop has both hardware and software.

Explain why the laptop needs both hardware and software.

(2)

(b) Zakir needs to choose between a solid state drive and a hard disc drive.

(i) Give **two** reasons, other than cost and capacity, why Zakir may choose a solid state drive rather than a hard disc drive.

(2)

1

2

(ii) Solid state drives store data on flash memory, which is similar to RAM.

Explain **one** difference between flash memory and RAM.

(2)

(c) One of the laptops has 4 megabytes (MB) of cache memory.

Explain the purpose of cache memory.

(2)

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3. Question 3 · part b.i · 2 marks · Working and answer

Working and answer

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4. Question 5 · part d.i · 2 marks

2022 JUN · Paper 01

Further practice · Attempt part d.i only.

5 Viza Health Centre is located in the North East of England.

(a) The health centre uses artificial intelligence to provide a symptom-checking service for its patients.

Patients log on to the website and input their symptoms.

(i) Describe how artificial intelligence could identify what is wrong with them.

(2)

(ii) Give **one** reason why a patient may not want to use this online service.

(1)

(b) The health centre has clinics in two buildings: Cleveland and Stockton.

The network server is in the Cleveland building.

(i) Name the type of network used to access the server from within the Cleveland building.

(1)

(ii) Name the type of network used to access the server from the Stockton building.

(1)

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4. Question 5 · part d.i · 2 marks

2022 JUN · Paper 01

Further practice · Attempt part d.i only.

(c) The network is at risk of an eavesdropping attack.

Identify the description of eavesdropping.

(1)

- ☒ **A** Tricking people into giving information by sending emails pretending to be from someone in authority
- ☒ **B** Spying on someone using a computer
- ☒ **C** Intercepting information as it is transmitted over a network
- ☒ **D** Redirecting a user from a genuine website to a fake one

(d) Doctors use laptops when they visit patients in their homes.

(i) The laptops have solid state drives.

Explain **one** reason why a solid state drive is better than a magnetic hard drive for the laptops.

(2)

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(ii) Describe how data is stored on a solid state drive.

(2)

(iii) The laptops have two types of memory.

Complete the table by adding **one** tick (✓) to match **each** description to the type of memory used.

(2)

Description	RAM	ROM
Stores the boot up sequence		
The contents are lost when the laptop is shut down		

(Total for Question 5 = 12 marks)

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4. Question 5 · part d.i · 2 marks · Working and answer

Working and answer

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5. Question 5 · part d.ii · 2 marks

2022 JUN · Paper 01

Further practice · Attempt part d.ii only.

5 Viza Health Centre is located in the North East of England.

(a) The health centre uses artificial intelligence to provide a symptom-checking service for its patients.

Patients log on to the website and input their symptoms.

(i) Describe how artificial intelligence could identify what is wrong with them.

(2)

(ii) Give **one** reason why a patient may not want to use this online service.

(1)

(b) The health centre has clinics in two buildings: Cleveland and Stockton.

The network server is in the Cleveland building.

(i) Name the type of network used to access the server from within the Cleveland building.

(1)

(ii) Name the type of network used to access the server from the Stockton building.

(1)

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5. Question 5 · part d.ii · 2 marks

2022 JUN · Paper 01

Further practice · Attempt part d.ii only.

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(c) The network is at risk of an eavesdropping attack.

Identify the description of eavesdropping.

(1)

- ☒ **A** Tricking people into giving information by sending emails pretending to be from someone in authority
- ☒ **B** Spying on someone using a computer
- ☒ **C** Intercepting information as it is transmitted over a network
- ☒ **D** Redirecting a user from a genuine website to a fake one

(d) Doctors use laptops when they visit patients in their homes.

(i) The laptops have solid state drives.

Explain **one** reason why a solid state drive is better than a magnetic hard drive for the laptops.

(2)

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(ii) Describe how data is stored on a solid state drive.

(2)

(iii) The laptops have two types of memory.

Complete the table by adding **one** tick (✓) to match **each** description to the type of memory used.

(2)

Description	RAM	ROM
Stores the boot up sequence		
The contents are lost when the laptop is shut down		

(Total for Question 5 = 12 marks)

5. Question 5 · part d.ii · 2 marks · Working and answer

Working and answer

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